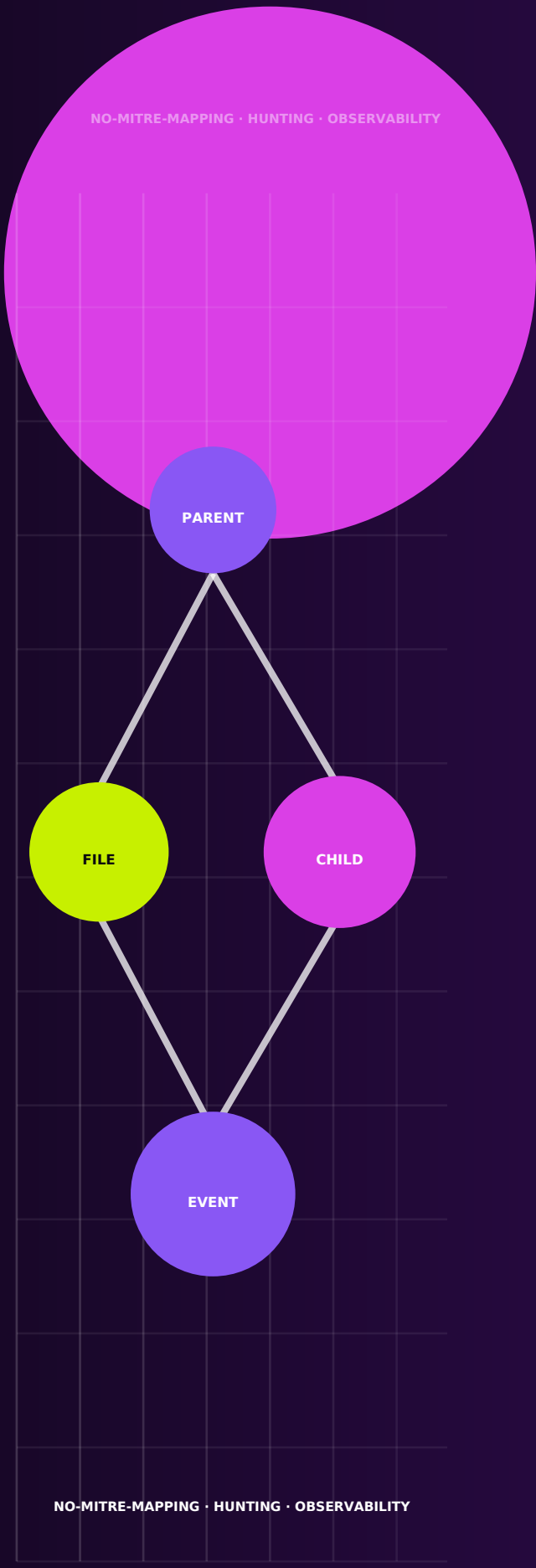
DETECTION THREAT LANDSCAPE

STUDIO 5000 ACD PATH TRAVERSAL - STUDIO 5000 NOVEL WRITE PATH

Baseline-assisted hunt for file-write consequences compatible with project path traversal.



HUNTING QUERY

CONFIDENCE: MEDIUM

REPORT TYPE
campaign

STATUS
Hunting / validation query - not exploit proof

PRIMARY PLATFORM
Microsoft Defender XDR / Microsoft Sentinel with MDE...

OPERATIONAL VALUE
9 / 10

01

Executive Assessment

Decision-grade summary with explicit evidence boundaries.

ASSESSMENT

DECISION

Retain this package as a hunting query and migrate it to the canonical visual and evidence contract without changing query conditions. Treat every result as an investigative lead until local schema, field population, benign overlap and the legacy claim provenance are reviewed.

LEGACY EVIDENCE RECORD

- The retained legacy source ledger contains 3 normalized references; claim-to-source mapping remains...

ANALYST ASSESSMENT

- The existing logic is retained at hunting maturity; repository migration is not execution eviden...
- The folder and query header map the package to NO-MITRE-MAPPING; operational relevance still depend...

DETECTION ASSUMPTIONS

- The named telemetry is collected across the intended asset or identity scope with adequate...
- Projected fields preserve their expected semantics in the target connector and parser. [C002]
- Approved activity, owners and change records are available to analysts during disposition. [C003]

KEY FINDINGS

FINDING 1

The original query conditions and source ledger were preserved during layout migration; legacy PDF SHA-256 07b4e07456b57955... is recorded for traceability. [C001]

FINDING 2

The principal observable is Evaluate DeviceFileEvents for studio 5000 novel write path using the existing source-derived conditions while preserving projected investigation context. [C002]

FINDING 3

The evidence boundary remains explicit: a match does not prove crafted content, project opening, path traversal, code execution or process-safety impact. [C003]

EVIDENCE BOUNDARY

A matching result does not prove crafted content, project opening, path traversal, code execution or process-safety impact. Target-environment field population, benign prevalence, parser behavior and operational ownership have not been validated by this repository migration.

02

Threat Mechanics

Technical mechanism, observable transitions and supported scope.

ATTACK FLOW

TECHNICAL BACKGROUND

The retained dossier links Studio 5000 Acd Path Traversal to the detection opportunity described as studio 5000 novel write path. The v2 migration preserves the query exactly and exposes its match boundary, but does not reinterpret legacy source claims as laboratory evidence. Evaluate DeviceFileEvents for studio 5000 novel write path using the existing source-derived conditions while preserving projected investigation context. [C001][C002]

1

THREAT CONTEXT

Baseline-assisted hunt for file-write consequences compatible with project path traversal. Observable: Context retained from the legacy dossier and source ledger.

2

OBSERVABLE CANDIDATE

Evaluate DeviceFileEvents for studio 5000 novel write path using the existing source-derived conditions while preserving projected investigation context. Observable: Events from DeviceFileEvents and projected fields.

3

CONTEXTUAL ENRICHMENT

Resolve asset, identity, process, ownership and change context before disposition. Observable: Raw event, surrounding timeline, inventory and authorized-change evidence.

4

DECISION GATE

Escalate only when corroborating evidence closes the documented non-claim boundary. Observable: Controlled validation, explainable negatives and analyst-readable evidence.

03

Telemetry & Evidence Model

Minimum data contract and known visibility gaps.

DATA CONTRACT

PRIMARY PLATFORM

Microsoft Defender XDR / Microsoft Sentinel with MDE tables. Evaluate DeviceFileEvents for studio 5000 novel write path using the existing source-derived conditions while preserving projected investigation context.

FIELD / TABLE	STATUS	PURPOSE
DeviceFileEvents	**Required**	Primary event source selected by the unchanged query logic.
Event time and entity identifiers	**Required**	Chronology, deduplication, scoping and investigation pivots.
Raw projected context	**Required**	Preserve the fields emitted by the query before normalization or tuning.
Asset, identity and change inventory	**Enrichment**	Separate authorized activity and define operational ownership.

KNOWN VISIBILITY GAPS

Unvalidated local schema Missing projected fields Incomplete retention or asset scope Mutable attacker behavior Legacy claim provenance review pending

04

Primary Detection Logic

Native query, explicit condition evidence and match meaning.

KQL

```
// Rule: [NO-MITRE-MAPPING] Studio 5000 Acd Path Traversal - Studio 5000 Novel Write Path
// MITRE ATT&CK: NO-MITRE-MAPPING
// Classification: Hunting query
// Status: Hunting / validation query - not exploit proof
// Primary platform: Microsoft Defender XDR / Microsoft Sentinel with MDE tables
// Required telemetry: DeviceFileEvents
// Purpose: Baseline-assisted hunt for file-write consequences compatible with project path traversal.
// Logic: Evaluate DeviceFileEvents for studio 5000 novel write path using the existing source-derived conditions while preserving projected investigation context.
// A match proves: A match indicates a baseline deviation compatible with the advisory's file-write consequence.
// A match does not prove: crafted content, project opening, path traversal, code execution or process-safety impact.
// Expected false positives: authorized administration, approved software, testing, or benign artifact collisions.
// Coverage limits: local schema, field population, retention, collection scope, and attacker variation require validation.
// Evidence claims: C001, C002, C003
//
// Title: Studio 5000 ACD-Associated Novel File Writes
// Status: Hunting / validation query - not exploit proof
// Confidence: Medium
// Platform: Microsoft Defender XDR / Microsoft Sentinel with MDE tables
// Telemetry: DeviceFileEvents
// Purpose: Correlate recent .ACD project activity with Rockwell-attributed file
// writes into device folders absent from the preceding 30-day baseline.
// Vulnerability: CVE-2026-9108
// Important: A match is only compatible with a path-traversal consequence. It
// cannot prove that Studio 5000 opened a crafted project or that exploitation
// occurred.

let DetectionWindow = 1d;
let BaselineWindow = 30d;
let BaselinePaths = materialize(
    DeviceFileEvents
    | where Timestamp between (ago(BaselineWindow) .. ago(DetectionWindow))
    | where InitiatingProcessVersionInfoCompanyName has "Rockwell"
        or InitiatingProcessVersionInfoProductName has "Studio 5000"
    | extend NormalizedFolder = tolower(FolderPath)
    | summarize by DeviceId, NormalizedFolder
);
let RecentAcd = materialize(
    DeviceFileEvents
    | where Timestamp >= ago(DetectionWindow)
    | where FileName endswith ".acd"
    | project DeviceId, DeviceName, AcdTime=Timestamp,
              AcdFileName=FileName, AcdFolderPath=FolderPath,
              FileOriginUrl, FileOriginReferrerUrl, FileOriginIP, SHA1
);
let NovelRockwellWrites =
    DeviceFileEvents
    | where Timestamp >= ago(DetectionWindow)
    | where ActionType in~ ("FileCreated", "FileModified", "FileRenamed")
    | where InitiatingProcessVersionInfoCompanyName has "Rockwell"
        or InitiatingProcessVersionInfoProductName has "Studio 5000"
    | extend NormalizedFolder = tolower(FolderPath)
// ... complete 68-line query in hunting.kql
```

WHAT A MATCH PROVES

A match indicates a baseline deviation compatible with the advisory's file-write consequence.

WHAT IT DOES NOT PROVE

crafted content, project opening, path traversal, code execution or process-safety impact.

MATURITY

Hunting query - Hunting / validation query - not exploit proof. Evaluate DeviceFileEvents for studio 5000 novel write path using the existing source-derived conditions while preserving projected investigation context.

05

Triage & Hunting Playbook

Evidence-preserving workflow from match to disposition.

WORKFLOW

01

Preserve the complete raw matching event, query version, timestamp, connector or sensor version and stable entity identifiers.

02

Confirm the event originated from the intended Microsoft Defender XDR / Microsoft Sentinel with MDE tables data source and verify the semantics of every condition used by the query.

03

Resolve asset, identity, process, software and destination ownership against authoritative inventories and approved change records.

04

Build the surrounding endpoint, identity, network or appliance timeline and seek an independent evidence layer before escalation.

05

Classify benign overlap explicitly; tune only with durable ownership or provenance controls, never by suppressing the core behavior globally.

06

Escalate only when corroborating evidence establishes malicious or unauthorized activity beyond the documented match boundary.

DISPOSITION RULE

Escalate only when the evidence establishes the behavior described by the query. crafted content, project opening, path traversal, code execution or process-safety impact.

06

False Positives & Validation

Overlap model, controlled checks and promotion gate.

QUALITY GATE

AUTHORIZED ADMINISTRATION OR SOFTWARE

Require owner, purpose, approved change and expected execution or configuration provenance.

BENIGN ARTIFACT OR COMMAND COLLISION

Require the full condition set plus an independent evidence layer or stable provenance.

PARSER, INVENTORY OR RETENTION AMBIGUITY

Reproduce a controlled event and reconcile raw fields, collection scope and authoritative inventory.

VALIDATION SEQUENCE

- Generate an authorized controlled equivalent of the primary observable without using live malicious infrastructure.
- Verify each query condition and projected field against retained raw telemetry.
- Run representative approved and near-match negative controls.
- Measure historical prevalence and classify every returned owner or workflow.
- Record parser, connector, retention and asset-scope dependencies with the test evidence.

SUCCESS

The controlled positive produces a complete, explainable row and representative negatives are separable through durable context.

REJECTION

Reject promotion when field semantics, collection scope, benign ownership or the core behavior cannot be established reproducibly.

PROMOTION GATE

Require claim-level source review, a retained controlled positive, representative negatives, measured prevalence and an accountable operational owner.

07

Analyst Assessment & References

Coverage statement, open questions and source record.

FINAL

CONCLUSION

This package now uses the canonical v2 layout and evidence contract while preserving the original hunting.kql conditions. It remains a hunting query at medium confidence. The visual migration is complete; claim-level provenance and target-environment execution remain explicit promotion gates.

MITRE ATT&CK

NO-MITRE-MAPPING

PRIMARY AND AUTHORITATIVE REFERENCES

S001 rockwellautomation.com - Security Advisories - CVE-2026-9108 (undated)

hxxps://www[.]rockwellautomation[.]com/en-us/trust-center/security-advisories[.]html

S002 nvd.nist.gov - National Vulnerability Database (undated)

hxxps://nvd[.]nist[.]gov/vuln/detail/CVE-2026-9108

S003 Microsoft Learn - DeviceFileEvents table in the advanced hunting schema (undated)

hxxps://learn[.]microsoft[.]com/en-us/defender-xdr/advanced-hunting-devicefileevents-table

OPEN QUESTIONS

A matching result does not prove crafted content, project opening, path traversal, code execution or process-safety impact. Target-environment field population, benign prevalence, parser behavior and operational ownership have not been validated by this repository migration.

DOCUMENT CONTROL

Schema 2.0 | Layout 2.0 | Maturity hunting | Confidence Medium | No customer data | Evidence ledger required